WEEK 4 · LECTURE L2 · UNIT II

E-Recruitment and Applicant Tracking

The hiring funnel digitized, the data it creates, and who it quietly screens out.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

- Describe the digitized hiring funnel and the data an ATS creates.
- Explain the main failure modes of applicant tracking systems.
- Apply selection validity research, disparate impact doctrine and realistic job previews.
- Evaluate whether an algorithmic screen is fair, using rates rather than totals.

> **COURSE OUTCOMES**
> CLO 2 (e-recruitment and ATS) and CLO 3 (fairness and bias).

THE FUNNEL, DIGITIZED

Where applicants leave the process

Stage	What the ATS records	What to check
Applications	Source channel, CV text, form data	Who never applies: channel reach
Screening	Keyword or model scores, pass or fail	The most automated stage; often removes half or more
Shortlist and interview	Recruiter actions, scheduling	Whether the interview actually discriminates
Offer and joining	Offers made, accepted, joined	Offer acceptance and early attrition

EXAM LENS

Analyze each stage with rates split by group, not with totals. Totals hide unfairness.

FAILURE MODES

How an ATS screens people out

Keyword screening

Rejects synonyms, regional job titles and CVs written in another language.

Career-gap penalties

Filters out returners, often women after caregiving, and seasonal workers.

Candidate experience

Long forms that fail on a phone and no status updates; applicants quietly give up.

An ATS automates the funnel, and any bias in its rules, at the same speed.

A WARNING

The same pass rate can hide different fairness

The trap

Two groups of requisitions, with and without an algorithm, can show almost identical overall screening pass rates.

Where the gap appears

Only when offers are split by gender and selection rates compared does an adverse impact become visible.

The test

Compare selection rates by group with the four fifths rule (Week 7); never compare raw counts.

CHANNELS

Channel choice is a design choice

Reach

Campus and professional networks tend to reach metro candidates; walk-ins and referrals reach local and frontline candidates.

Cost and speed

Cost per hire and days to fill differ by channel; compare both, with quality of hire.

Fairness

A channel that reaches only some groups produces an unequal pool before any screening happens.

THEORY IN DEPTH · SCHMIDT AND HUNTER (1998); SACKETT ET AL. (2022)

How well do selection methods predict performance?

THE CORE IDEA

Validity is the correlation between a selection score and later job performance.

HOW IT WORKS

Schmidt and Hunter estimated general mental ability and structured interviews at about .51; Sackett et al. revised general mental ability down to about .31.

LIMITS AND CRITIQUE

In the revised estimates, structured interviews (about .42) are the strongest single predictor.

IN PRACTICE

A keyword screen usually has unknown validity; ask the vendor for validation evidence before trusting it.

EXAM LENS

Cite the 2022 revision: it shows evidence evolves, and corrects a widely repeated claim.

Results count, not intentions

THE CORE IDEA

The US Supreme Court held that neutral-looking tests are unlawful if they exclude groups and are not job related.

HOW IT WORKS

This disparate impact doctrine led to the 1978 Uniform Guidelines and the four fifths rule.

LIMITS AND CRITIQUE

The four fifths rule is a rule of thumb, not a statistical test; small samples can mislead.

IN PRACTICE

An algorithm need not intend bias; what matters is whether its outcomes differ by group without job-related justification.

EXAM LENS

Note that this is US law; Indian law has no direct equivalent, but the analytical logic of disparate impact still applies.

THEORY IN DEPTH · DASTIN (2018): THE AMAZON CASE

A hiring model can learn past bias

THE CORE IDEA

Amazon dropped an experimental resume-scoring tool after it was found to downgrade resumes mentioning women's activities.

HOW IT WORKS

Trained on a decade of mostly male hires, it learned to treat male-linked signals as merit.

LIMITS AND CRITIQUE

Removing gender words did not fix it, because other features acted as proxies.

IN PRACTICE

Any vendor ATS was trained on someone's past data; ask whose, and audit its outcomes by group.

EXAM LENS

Use the case to illustrate proxy discrimination, then link it to Barocas and Selbst's routes (Week 5).

THEORY IN DEPTH · WANOUS (1973); PREMACK AND WANOUS (1985)

Telling the truth about a job improves retention

THE CORE IDEA

A realistic job preview shows both the good and the hard parts of a job before the offer.

HOW IT WORKS

Candidates self-select out early, and those who join have accurate expectations.

LIMITS AND CRITIQUE

The effect on turnover is real but modest in meta-analysis.

IN PRACTICE

For high-attrition frontline roles, a short phone video of a real shift may do more than another screening filter.

EXAM LENS

Report the meta-analytic effect honestly: modest, but reliable.

PRACTICE FOR THE END-TERM EXAMINATION

Questions to rehearse

"An applicant tracking system automates bias at scale." Critically evaluate with reference to the Amazon case, disparate impact doctrine and selection validity research. **20 marks**

A strong answer: explains failure modes; uses Griggs and the four fifths logic; distinguishes validity from fairness; proposes audits and design safeguards.

Discuss the role of realistic job previews and channel choice in reducing early attrition among frontline hires. **20 marks**

A strong answer: explains the self-selection and met-expectations mechanisms; reports meta-analytic evidence; links channels to reach and fairness.

What did Sackett et al. (2022) change about our understanding of selection validity? **10 marks**

A strong answer: gives the revised estimates and their implication for structured interviews.

CARRY THESE INTO THE EXAMINATION

Three things to remember

1. **An ATS digitizes the funnel, and the biases of whoever built its rules**

2. **Check every stage with rates split by group**
   Totals hide unfairness.

3. **Validity and fairness are different questions**
   A screen can be fast, valid and still unfair; test both.

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- An Indian HRIS or ATS implementation case (shared in class).

COMPANION

Each set reading is explained in the matching Unit Reader on rksingh.tech/materials.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

Works named on the slides of this lecture. They are background sources, not set readings.

Dastin, J. (2018, October 10). Amazon scraps secret AI recruiting tool that showed bias against women. *Reuters*.

Griggs v. Duke Power Co., 401 U.S. 424 (1971).

Premack, S. L., & Wanous, J. P. (1985). A meta-analysis of realistic job preview experiments. *Journal of Applied Psychology, 70*(4), 706–719. https://doi.org/10.1037/0021-9010.70.4.706

Sackett, P. R., Zhang, C., Berry, C. M., & Lievens, F. (2022). Revisiting meta-analytic estimates of validity in personnel selection: Addressing systematic overcorrection for restriction of range. *Journal of Applied Psychology, 107*(11), 2040–2068. https://doi.org/10.1037/apl0000994

Schmidt, F. L., & Hunter, J. E. (1998). The validity and utility of selection methods in personnel psychology: Practical and theoretical implications of 85 years of research findings. *Psychological Bulletin, 124*(2), 262–274. https://doi.org/10.1037/0033-2909.124.2.262

Wanous, J. P. (1973). Effects of a realistic job preview on job acceptance, job attitudes, and job survival. *Journal of Applied Psychology, 58*(3), 327–332. https://doi.org/10.1037/h0036305